

Moduli of Coble Surfaces

D. Zack Garza

NCTS

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- ① Moduli of K3 Surfaces
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Introduction

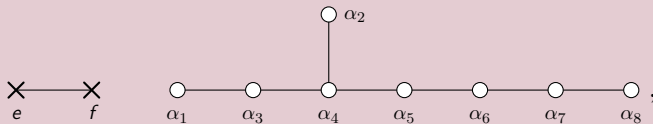
- K3 surface: $q(X) := h^1(\mathcal{O}_X) = 0$ and $\omega_X \cong \mathcal{O}_X$.
- Enriques surface: $q(X) = 0$ and $\omega_X \not\cong \mathcal{O}_X$ but $\omega_X^2 \cong \mathcal{O}_X$.
- Coble surface: $q(X) = 0$, $h^0(\omega_X^{-1}) = 0$ but $h^0(\omega_X^{-2}) > 0$.

Surface	$\text{ord}(K_X)$	$\kappa(X)$	$\text{Pic}(X)/\text{tors}$	$h^{2,0}$	$h^{1,1}$	$h^{0,2}$
K3	1	0	$L_{K3} := U^3 \oplus E_8^2$	1	20	1
Enriques	2	0	$L_{En} := U \oplus E_8$	0	10	0
Coble	∞	$-\infty$	$L_{Co} := \langle -1 \rangle \oplus U \oplus E_8$	0	11	0

Here $\langle -1 \rangle$ is generated by K_X where $K_X^2 = -1$.

Lattices

$$G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad G_{E_8} = \begin{pmatrix} -2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -2 \end{pmatrix}.$$



$$\times : v^2 = 0, \quad \circ : v^2 = -2, \quad \bullet : v^2 = -4$$

Nonsymplectic Involutions

If $G \curvearrowright L$, define

$$L^G := \left\{ v \in L \mid G.v = v \right\} \leq L, \quad L_G := (L^G)^{\perp L}$$

- Y Enriques $\implies \exists X \rightarrow Y$ its universal cover by a K3.
- $\pi_1(Y) = \langle \iota_{\text{En}} \rangle \cong \mathbf{Z}/2\mathbf{Z}$ is a **nonsymplectic involution**:

$$\iota_{\text{En}}(\sigma_X) = -\sigma_X \quad \text{where} \quad H^{2,0}(X) = \mathbf{C}\sigma_X.$$

- Yields

$$S_{\text{En}} := L_{K3}^{\langle \iota_{\text{En}} \rangle} \cong L_{\text{En}}(2) = U(2) \oplus E_8(2).$$

- [Nik79a]: if $\iota \curvearrowright X$ is a nonsymplectic involution on a K3 surface, there are 75 possibilities for

$$S(\iota) := L_{K3}^{\langle \iota \rangle} \quad \text{and} \quad T(\iota) := S^{\perp L_{K3}}.$$

Nikulin's Pyramid

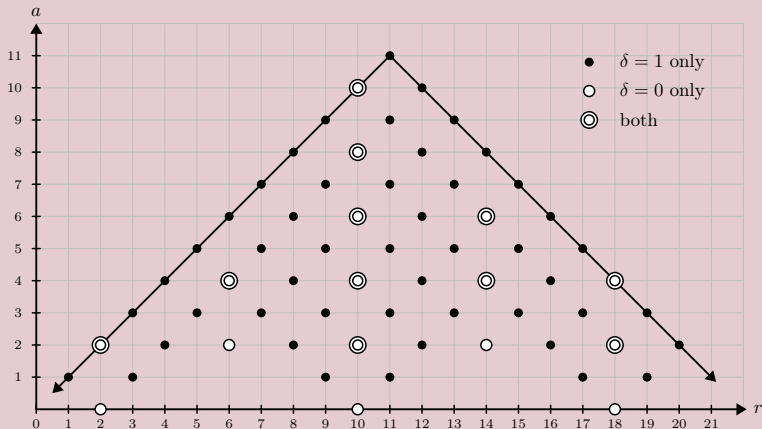


Figure: $S_{dP} = (2, 2, 0)$, $S_{En} = (10, 10, 0)$, $S_{Co} = (11, 11, 1)$.

Period Domains

For any lattice T of signature $(2, n)$, there is a *period domain* classifying Hodge structures of type T , a connected component of

$$\Omega_T := \left\{ v \in \mathbf{P}(T \otimes_{\mathbf{Z}} \mathbf{C}) \mid v^2 = 0, \bar{v} \cdot v > 0 \right\} = D_T \amalg \bar{D}_T.$$

Always a Type IV Hermitian symmetric domain. When T is of geometric origin¹ and $\Gamma_T \leq \mathrm{O}(T)$ the monodromy group of the corresponding ZVHS, $\exists$ a moduli space S -polarized K3 surfaces

$$F_T := \Gamma_T \backslash D_T.$$

¹e.g. $T = S^{\perp L_{K3}}$ for $S \hookrightarrow \mathrm{Pic}(X)$ with X a K3 surface.

TBD

Examples:

$\mathcal{M}$	S	T	S'	T'	Γ_T
F_{2d}	$\langle 2d \rangle$	$\langle -2d \rangle \oplus U^2 \oplus E_8^2$	$I_{1,0}(2d)$	$I_{0,1}(2d) \oplus II_{2,18}$	$O^+(T) \cap \tilde{O}(T)$
$F_{\text{ell}}[\text{ABE22}]$	U	$U^2 \oplus E_8^2$	$II_{1,1}$	$II_{2,18}$	$O^+(T) \cap \tilde{O}(T)$
$F_{(2,2,0)}[\text{AE22}]$	$U(2)$	$U \oplus U(2) \oplus E_8^2$	$II_{1,1}(2)$	$II_{1,1}(2) \oplus II_{1,17}$	$O^+(T) \cap \tilde{O}(T)$
$F_{\text{En}}[\text{Ste95}]$	$U(2) \oplus E_8(2)$	$U \oplus U(2) \oplus E_8(2)$	$II_{1,9}(2)$	$II_{1,1} \oplus II_{1,9}(2)$	$O^+(T) \cap \tilde{O}(T)$
$F_{\text{En},h}[\text{Ste95}]$	$\dots$	$\dots$	$\dots$	$\dots$	$O^+(T) \cap \Gamma_{T,h}$

Where

$$\rho_T : O(T) \rightarrow O(A_T), \quad h := e + f \in S_{\text{En}} = U(2) \oplus E_8(2),$$

$$\Gamma_{T,h} = \rho_T^{-1}(\rho_S(\text{Stab}_{O(S)}(h))), \quad \tilde{O}(T) := \rho_T^{-1}(0).$$

(N.B.: F_{2d} due to [Sca87, Laz16, AET23].)

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Compactifications

By [BB66], F_T is a quasiprojective variety. When $\mathcal{M}$ is a moduli problem with coarse space F_T , there are many compactifications:

$$\overline{\mathcal{M}}^{\text{GIT}}, \quad \overline{F}_T^{\text{BB}}, \quad \overline{F}_T^{\Sigma^\bullet}, \quad \overline{F}_T^{\mathcal{F}^\bullet}, \quad \overline{\mathcal{M}}^{\text{KSBA}}.$$

For the KSBA compactifications, take pairs $(X, \varepsilon R)$ where

F_{2d}	$(X, L), L^2 = 2d$	$R^{\text{rc}} = \left(\sum_{\mathbf{P}^1 \cong C \in L } n_C C \right) \in n_d L $
F_{ell}	$(X, s + (d+1)f)$	$R^{\text{Ram}} := \text{Ram}(X \xrightarrow{2\text{-to-}1} \mathbf{P}(1, 1, 4))$
$\dots$	$\dots$	$R^{\text{rc}} \approx s + \sum_{i=1}^{24} f_i$
$F_{(2,2,0)}$	(X, L)	$R^{\text{Ram}} := \text{Ram}(\iota_{\text{dP}} \curvearrowright X) \text{ with } S = U(2)$
$F_{\text{En},2}$	$(Z, [L_Z])$	$R^{\text{Ram}} := \text{Ram}(Z \xrightarrow{2\text{-to-}1} \text{dP}_4)$

Theorem ([AE22, Thm. 9.10])

In 50 out of 75 cases for T , one has

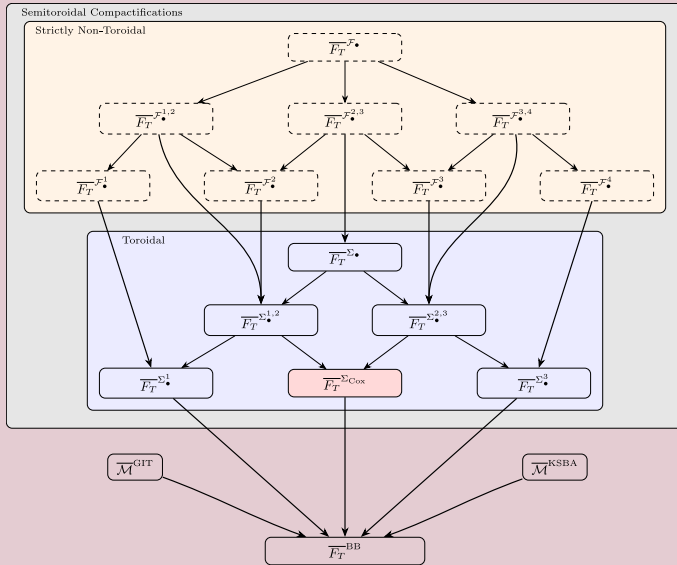
$$\overline{F}_T^{\mathcal{F}_\bullet(\Phi_{2,4})} \rightarrow \overline{F}_T^{\mathcal{F}_\bullet(\Phi_2)} \rightarrow \overline{F}_T^{\mathcal{F}_\bullet^{\text{ram}}} \xleftarrow{\sim} (\overline{F}_T^{\text{KSBA}})^\nu$$

where $\mathcal{F}_\bullet^{\text{ram}}$ is a collection of semifans indexed by $\partial_0 \overline{F}_T^{\text{BB}}$.

$$\Phi(L) = \left\{ \alpha \in L \mid s_\alpha \in \mathcal{O}(L) \right\}, \quad \Phi_{2,4}(L) := \left\{ \alpha \in \Phi(L) \mid \alpha^2 = -2, -4 \right\}$$

$$s_\alpha(v) = v - (2v\alpha)\hat{\alpha}, \quad \hat{\alpha} := \alpha/\alpha^2$$

Stratified Spaces and Posets



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Coble Surfaces

Definition (Coble Surfaces)

A **Coble surface** [Cob19, Oda85, DZ99, ?] is a smooth, projective, rational surface Y with

$$h^0(\omega_Y^{-1}) = 0, \quad h^0(\omega_Y^{-2}) \neq 0 \quad (\iff | -K_Y | = \emptyset, \quad | -2K_Y | \neq \emptyset).$$

It is *terminal of K3 type* if $| -2K_Y | \ni C = \sum_{i=1}^n C_i$ with $C_i \cong \mathbf{P}^1$ and $C_i C_j = -4\delta_{i=j}$. Write $F_{\text{Co}} := \{\text{Coble surfaces } Y\} / \sim$.

Fact

Every Coble is a basic rational surface: $\rho : Y \rightarrow \mathbf{P}^2$ decomposes as a sequence of blowups at $N = 9 + n$ points, and

$$\rho(C) \in | -2K_{\mathbf{P}^2} | = | \mathcal{O}_{\mathbf{P}^2}(6) |.$$

Why Cobles are of interest

In [AEGS23] we construct $\overline{F_{\text{En},2}} \cong \overline{D_{T_{\text{En}}}/\Gamma_{T_{\text{En},2}}}$. By [Nam85],

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$$T_{\text{En}}[-2]/\Gamma_{T_{\text{En},2}} = \{\alpha\}, \quad \Delta_{-2} = \alpha^\perp$$

are Coble surfaces with a $\frac{(1,1)}{4}$ singularity,

-

$$T_{\text{En}}[-4]/\Gamma_{T_{\text{En},2}} = \{\beta_1, \beta_2\}, \quad \Delta_{-4} = \beta_1^\perp \cup \beta_2^\perp,$$

- $\beta_1^\perp$ are *nodal* Enriques surfaces (resolution contains a (-2) -curve),
- $\beta_2^\perp$ are *unigonal* Enriques surfaces $Z \xrightarrow{2\text{-to-1}} \mathbf{P}(1, 1, 2)$.

So $F_{\text{Co}} \hookrightarrow F_{\text{En},2}$ as a divisor, as do F_{Nod} and F_{uni} . So there exist KSBA compactifications of all three divisors.

Coble Surfaces

We primarily consider the $n = 1$ case, so $K_Y^2 = -1$ and there are $N = 10$ blowups.

Construction

Let $C \subseteq \mathbf{P}^2$ be an irreducible rational sextic, then

$$p_a(C) = \frac{(d-1)(d-2)}{2} + \sum_{p \in C^{\text{sing}}} \delta_p \implies \sum_p \delta_p = 10.$$

Choose C with at worst slc singularities, then $C \in S_{6,0}$ generically satisfies $C^{\text{sing}} = A_1^{10}$ but admits many singularity profiles:

$$\delta(A_k) = \lceil k/2 \rceil \rightsquigarrow A_{20}, A_1^{10}, A_1^9 + A_2, A_1^8 + A_3, A_7 + A_6, \\ D_4 + A_1^7, E_6 + A_1^7, E_8 + A_1^6, \dots$$

Coble Surfaces

Then take $\rho : Y := \text{Bl}_{\text{Csing}}(\mathbf{P}^2) \rightarrow \mathbf{P}^2$, and $\tilde{C} \in |-2K_Y|$ defines a 2-to-1 cover $\pi : X \rightarrow Y$ by a K3 surface X . We have

$$\text{Pic}(Y) = \text{Pic}(\mathbf{P}^2) \oplus \langle E_1, \dots, E_{10} \rangle \cong I_{1,10}, \quad \pi^* : \text{Pic}(Y)(2) \hookrightarrow L_{K3}$$

The image is a 2-elementary [Nik79b] lattice, thus

$$S_{\text{Co}} := \text{Pic}(Y)(2) \cong I_{1,10}(2) \cong \langle -2 \rangle \oplus U(2) \oplus E_8(2) = (11, 11, 11)_1$$

$$T_{\text{Co}} := S_{\text{Co}}^{\perp L_{K3}} \cong I_{2,9}(2) \cong \langle 2 \rangle \oplus U(2) \oplus E_8(2) = (11, 11, 11)_2$$

$$\implies \exists \iota_{\text{Co}} \hookrightarrow X \text{ nonsymplectic, } L_{K3}^{\langle \iota_{\text{Co}} \rangle} \cong S_{\text{Co}}, \quad L_{K3}^{\langle \iota_{\text{Co}} \rangle} \cong T_{\text{Co}}$$

$$\implies F_{\text{Co}} \hookrightarrow \overline{F_{I_{1,10}(2)}} = \overline{F_{(11,11,1)}} \text{ and } F_{\text{Co}} \hookrightarrow \overline{F_{\text{En},2}} \hookrightarrow \overline{F_{(2,2,0)}}.$$

Coble Surfaces

Note $F_{\text{Co}} \subseteq (\mathbf{P}^2)^{10} / \text{PGL}_3$, and [Cob19] shows there are 3 discriminant conditions, so $\dim F_{\text{Co}} = 12 - 3$ agrees with $\dim(F_{\text{I}_{1,10}(2)}) = 11 - 2$ and this is an open subset.

Goal

In [AEGS23], we show $\overline{F_{\text{En},2}}^\nu \cong \overline{F_{\text{En},2}}^{\mathcal{F}\bullet}$ for a specific semifan. Since $F_{\text{Co}} \hookrightarrow \overline{F_{\text{En},2}}$, its closure defines a KSBA compactification $\overline{F_{\text{Co}}}$, and one can hope to recover $\partial \overline{F_{\text{Co}}}$ from its intersection with $\partial \overline{F_{\text{En},2}}$:

Conjecture

There is an isomorphism $\overline{F_{\text{Co}}}^\nu \xrightarrow{\sim} \overline{F_{\text{Co}}}^{\mathcal{F}\bullet}$ where the semifan $\mathcal{F}$ can be made explicit, and stable models of degenerations of Coble surfaces can be effectively computed, giving an effective and complete description of $\partial \overline{F_{\text{Co}}}$.

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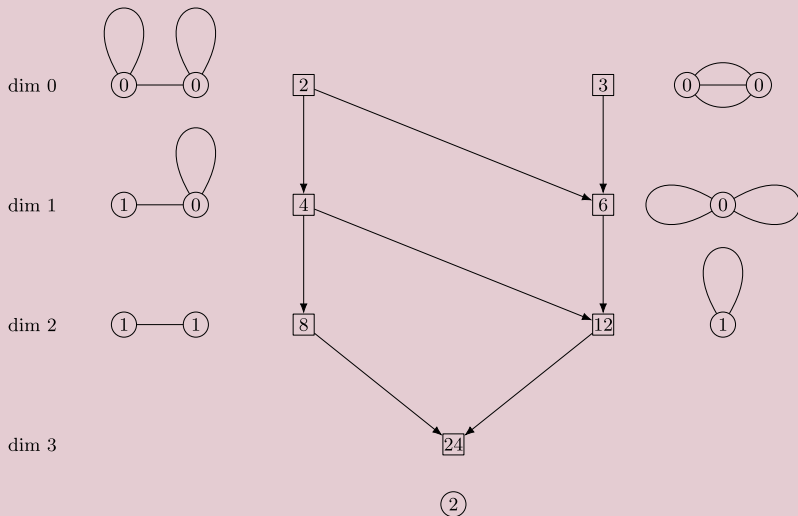
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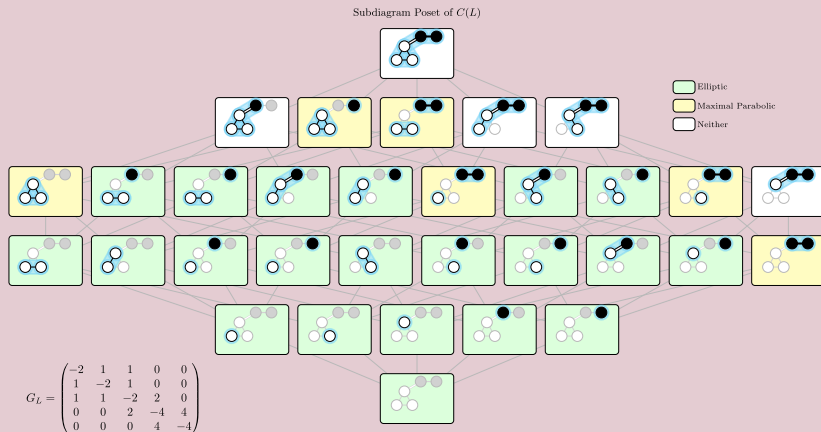
4 Boundaries

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$$\partial \overline{\mathcal{M}}_{2,0} \cong \text{Div}(24) \setminus \{1\} \cong \text{SubDiag}(\text{trivalent graph}) \subseteq I_{1,3}(2) \otimes_{\mathbf{Z}} \mathbf{R} - \{\emptyset\}$$



Stratification posets from Coxeter diagrams



Strategy

- Compute the polarization L_{Y_0} on a Coble central fiber Y_0 of a degeneration $(\mathcal{Z}, \mathcal{L}_{\mathcal{Z}})$ of Enriques surfaces in $F_{\text{En},2}$.
- $\exists!$ open immersions of period domains

$$F_{\text{Co}} \hookrightarrow F_{\text{I}_{1,10}(2)} \hookrightarrow F_{\text{En},2} \hookrightarrow F_{(2,2,0)}$$

- Compute the monodromy group Γ_{Co} and descend to quotients.
- Construct $\overline{F_{\text{Co}}}^{\text{BB}}$ and compute $\partial \overline{F_{\text{Co}}}^{\text{BB}}$.
- Compute the semifan refinement inducing $\overline{F_{\text{Co}}}^{\mathcal{F}\bullet} \rightarrow \overline{F_{\text{Co}}}^{\text{BB}}$ that coincides with $\overline{F_{\text{Co}}}$.
- Compute the induced divisors R_Y for induced stable pairs $(Y, \varepsilon R_Y)$ in the closure of F_{Co} in $\overline{F_{\text{En},2}}$ or $\overline{F_{(2,2,0)}}$.
- Compute the intersection of $\partial \overline{F_{\text{Co}}}$ with $\partial \overline{F_{\text{En},2}}$ lattice-theoretically.

Strategy

- Use the fibration $\overline{F}_{Co} \rightarrow \overline{F}_{Co}^{BB}$ to enumerate strata of $\partial\overline{F}_{Co}$.
- Construct stable models as slices/restrictions/etc of stable models for $\partial\overline{F}_{En,2}$, contracted from **integral affine** manifolds: $\mathbf{P}^1(\mathbf{R})$ or $\mathbb{D}$ folded from IAS^2 for K3 surfaces.

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5 Work In-Progress

Preliminary Results

Theorem

There are isometries

$$S_{\text{Co}} \cong I_{1,10}(2) \cong \langle -2 \rangle \oplus U(2) \oplus E_8(2) = (11, 11, 1)_1$$

$$T_{\text{Co}} \cong I_{2,9}(2) \cong \langle 2 \rangle \oplus U(2) \oplus E_8(2) = (11, 11, 1)_2$$

Theorem

There is a primitive embedding of lattices

$$T_{\text{Co}} \cong \langle 2 \rangle \oplus U(2) \oplus E_8(2) \hookrightarrow T_{\text{En}} = U \oplus U(2) \oplus E_8(2)$$
$$h \oplus u \oplus \alpha \mapsto (e + f) \oplus u \oplus \alpha$$

Preliminary Results

Theorem

The sequence of primitive embeddings

$$T_{\text{Co}} \hookrightarrow T_{\text{En}} \hookrightarrow T_{(2,2,0)} \hookrightarrow L_{K3}$$

is unique (as a sequence of embeddings) up to $O(L_{K3})$, and thus the corresponding maps on period domains are well-defined.

Theorem

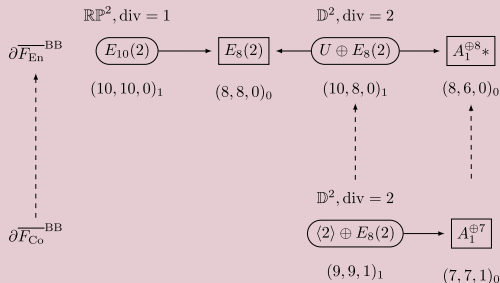
*The quotient $F_{\text{Co}} := D_{T_{\text{Co}}}/O^+(T_{\text{Co}})$ is a moduli space of **unpolarized** Coble surfaces, and there is a moduli space $F_{\text{Co},2} := D_{T_{\text{Co}}}/\Gamma_{T_{\text{Co},2}}$ of **polarized** Cobles surfaces. If the unique orbit δ in $T_{\text{En}}[2]/O^+(T_{\text{En}})$ remains unique in $T_{\text{En}}[2]/\Gamma_{\text{En},2}$, then there are identifications*

$$\Gamma_{T_{\text{Co},2}} \cong \text{im}(Z_{O(T_{\text{En}})}(\delta) \rightarrow O(\delta^\perp)) \text{ and } \delta^\perp \cong T_{\text{Co}}.$$

Coble Surfaces

Theorem

The morphism $\eta : F_{\text{Co}} \rightarrow F_{\text{En}}$ (unpolarized!) extends to an morphism $\partial \overline{F}_{\text{Co}}^{\text{BB}} \rightarrow \partial \overline{F}_{\text{En}}^{\text{BB}}$:



Why: by [KK72], this is because (1) both sides are *bounded* symmetric domains, (2) η is holomorphic, (3) both groups are arithmetic, (4) η is equivariant, and (5) HSDs are hyperbolically embedded in their Baily-Borel compactifications.

Coble Surfaces

Theorem

$F_{\text{Co},2}$ is the normalization of a closed subvariety of $F_{\text{En},2}$.

Idea: restrict $T_{\text{Co}}/\Gamma_{\text{Co},2} \rightarrow T_{\text{En}}/\Gamma_{\text{En},2}$ to scheme theoretic image
 $\implies$ birational. Proper and quasi-finite (because finite index subgroup)
 $\implies$ finite. Both are normal by [BB66]; apply Zariski's main theorem and identify image with a Noether-Lefschetz locus.

Conjecture

Let $X \rightarrow \tilde{Y}$ be the K3 cover of a Coble surface, with $Y = X/\iota_{\text{Co}}$. Then ι_{Co} is the limit of an Enriques involution ι_{En} which commutes with a del Pezzo involution $\iota_{\text{dP}} \curvearrowright X$. Thus $\iota_{\text{dP}} \curvearrowright Y$ yields an ample $\mathbf{Q}$ -Cartier ramification divisor. Resolving the $\frac{(1,1)}{4}$ singularity $Y \rightarrow \tilde{Y}$, the pair

$$(Y, \frac{1+\varepsilon}{2} C + \varepsilon \text{Ram}(\iota_{\text{dP}}))$$

is KSBA stable and induced by taking the closures

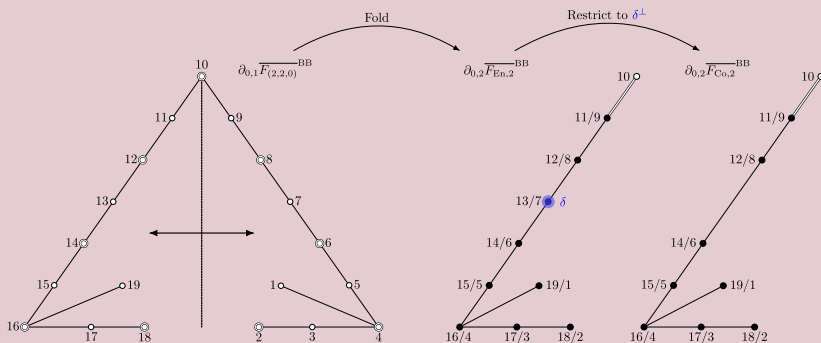
$$F_{\text{Co}} \hookrightarrow \overline{F_{\text{En},2}} \hookrightarrow \overline{F_{(2,2,0)}}.$$

Theorem

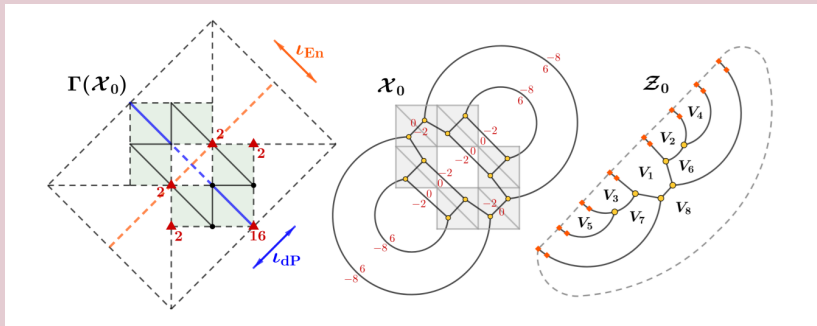
For the unique $v_0 \in T_{\text{Co}}[0]/O^+(T_{\text{Co}})$, there is an identification

$$v_0^\perp / v_0 \cong \langle 2 \rangle + E_8(2) \cong (9, 9, 1)_1$$

whose Coxeter diagram is obtained from $F_{\text{En},2}$ by restriction:



IAS² and dlt models

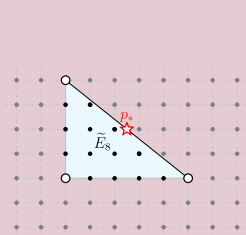
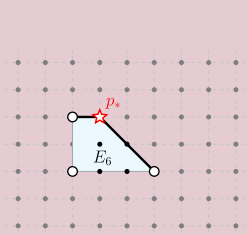
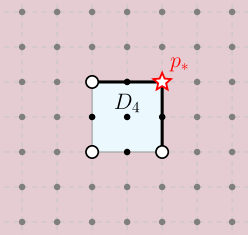
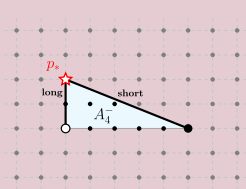
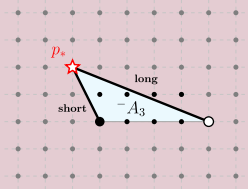
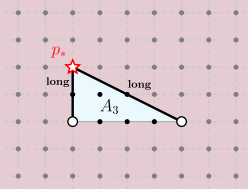


Surfaces appearing as (double covers of) irreducible components in divisorial log terminal models:

- $\text{Bl}_4 \text{Tor}(\mathbf{P}^1 \times \mathbf{P}^1)$
- $\text{Bl}_2 \text{Tor} \mathbf{F}_4$
- $\text{Bl}_{2\text{Nontor}}(\mathbf{P}^1 \times \mathbf{P}^1)$
- $\mathbf{F}_2$
- An A_{15} ADE surface

One takes the corresponding integral affine manifolds and performs contractions to obtain KSBA stable models.

Irreducible components of KSBA Stable Models



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